Yewon Jang

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EDUCATION

Hanyang University Seoul, South Korea
B.S. in Mechanical Engineering, GPA 3.61/4.0 (94.8/100) Mar 2022 – Aug 2026
Relevant Courses: Strength of Materials, Fluid Mechanics, Dynamics, Engineering Mathematics, Numerical Analysis, CAD
Ohio Northern University Ada, OH, USA
Exchange Program in Mechanical Engineering, Aug 2024 – May 2025
Relevant Courses: Finite Element Analysis

RESEARCH INTERESTS

Computational Mechanics: Developing physics-informed machine learning and surrogate models to overcome the computational costs and convergence bottlenecks of traditional finite element solvers.
Biomechanics: Extending patient-specific 3D modeling from linear-elastic toward hyper-elastic and contact-mechanics frameworks to capture complex material and interface behaviors more accurately.

PUBLICATION AND PRESENTATION

Yewon Jang, Colleen H. Kottmyer, Hui Shen, "A Finite Element Analysis Comparing the Biomechanical Stability of Fascia-Wrapped versus PRF-Embedded Diced Cartilage Grafts in Dorsal Augmentation Rhinoplasty", Proceedings of ASEE – NCS Conference, 2026, Alliance, OH

Cheng, X., Geibel, H. G., Clawson, S. J., **Jang, Y**., Hastings, T. C., Guerra, F. S., Bak, P. D., Yang, G., Herdman, V. T., Betts, C. M., and Shen, H., 2025, "A Photogrammetry-Based Approach for Patient-Specific Modeling in Syndactyly Surgery: Prototyping, 3D Reconstruction, and Finite Element Analysis," Proceedings of the ASME 2025 International Mechanical Engineering Congress and Exposition (IMECE2025), Memphis, TN, USA, Nov. 16–20, Paper No. IMECE2025-166019.

Yang, G., Shen, H., **Jang, Y**., and Cheng, X., "Finite Element Analysis-Assisted Surgical Planning and Evaluation of Flap Design in Hand Surgery," Frontiers in Bioengineering and Biotechnology, 13 (2025): 1611993

RESEARCH EXPERIENCE

Hanyang University & Ohio Northern University Seoul (Remote) & OH (In-person)
Undergraduate Research Assistant /Mechanical Engineering Aug 2024 – Jan 2026

Project: Biomechanical Analysis of Rhinoplasty Graft Stability
Evaluated the biomechanical stability of DC-F and DC-PRF grafts in rhinoplasty via FEA to quantify stress distributions and predict post-operative failure mechanisms.

- **Lead Researcher:** Spearheaded the end-to-end research workflow, from 3D modeling to manuscript preparation.
- **Advanced Modeling:** Developed 3D nasal models in ANSYS, applying the Rule of Mixtures to calculate effective elastic moduli for composite grafts.
- **Time-Variant Simulation:** Executed stage-based FEA, identifying 1-month post-op (T2) as the critical period for peak mechanical stress.
- **Clinical Insight:** Quantified the stress–stability trade-off between the two techniques- DC-F showed greater initial structural stability but 10–38× higher peak von Mises stress at one month post-op, while DC-PRF remained compliant but more load-type sensitive.

Project: Syndactyly Surgery Optimization via Photogrammetry & FEA
Contributed FEA modeling to a multi-institution project building a photogrammetry-based pipeline for patient-specific 3D hand reconstruction, applied to dorsal flap design in syndactyly repair.

- **Simulation Lead:** Performed the ANSYS structural analysis on hand models reconstructed by the team's photogrammetry system, including convergence checks and boundary-condition setup from patient anatomical data.
- **Flap Optimization:** Analyzed hexagonal-flap biomechanics during suturing, extracting patient-specific flap dimensions (h, w) from the reconstructed 3D hand geometry.
- **Convergence Solution:** Resolved 3D contact-convergence failures in complex web-space topographies by adapting a 3D-to-2D surface-unrolling method to ensure numerical stability.
- **Reliability Engineering:** Conducted Mesh Convergence Studies to validate the numerical stability of surgical models.

Project: Review of FEA Applications in Hand Surgery
Reviewed FEA-assisted surgical planning for hand skin flaps, highlighting current trends in patient-specific modeling and AR/MR integration.

- **Literature Synthesis:** Compiled material models, geometries, and boundary/loading conditions from FEA flap-design studies (2000–2025) into comparative tables.
- **Gap Identification:** Identified skin-property characterization and cross-study validation as the field's central limitations, informing the review's recommendation for AI-driven, adaptive flap-design tools.

ACADEMIC PROJECTS

Undergraduate Thesis: Multi-Linked Wheeled Snake Robot Spring 2025
Developed an 8-segment hybrid robot and a Python-based spatial path-following algorithm for high-precision trajectory tracking.

Braille Display Device for Korean Text Conversion Winter 2026
Designed a solenoid-actuated mechatronics system to convert real-time Korean text input into tactile Braille patterns.

Internal Combustion Engine Performance Analysis Spring 2024
Analyzed engine efficiency (IMEP, ISFC) and thermal performance based on Start of Injection (SOI) variables using P-V diagrams.

SCHOLARSHIPS

Scholarship	Institution	Year	Semester
Merit Based Scholarship	Hanyang University	2023	Spring Semester
National Scholarship (Co-op) Of Creative Korea 2	Hanyang University	2023	Spring Semester
Korea-U.S. Student Exchange Program in the field of high-tech industry	Korea Institute for Advancement of Technology (KIAT)	2024-2025	Fall Semester – Spring Semester

SKILLS

Technical: Catia, Ansys, SolidWorks, Python, MATLAB, Excel
GRE Q: 170, V: 151, AW: 3.5
TOEFL iBT: 5.0/6.0 (Reading: 4.5, Listening: 5.0, Speaking: 4.5, Writing: 5.5) | *Oct 2026*